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Electrically programmable fuse over crystalline semiconductor materials

Abstract

Embodiments of the disclosure provide an electrically programmable fuse (efuse) over crystalline semiconductor material. A structure according to the disclosure includes a plurality of crystalline semiconductor layers. Each crystalline semiconductor layer includes a compound material. A metallic layer is on the plurality of crystalline semiconductor layers. The metallic layer has a lower resistivity than an uppermost layer of the plurality of crystalline semiconductor layers. A pair of gate conductors is on respective portions of the metallic layer. The metallic layer defines an electrically programmable fuse (efuse) link between the gate conductors.

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Background/Summary

BACKGROUND

Field of the Invention

(1) The present invention relates to integrated circuits (ICs) and, more particularly, to electrically programmable fuses (efuses).

Description of Related Art

(2) Integrated circuit (ICs) incorporating III-V semiconductor materials, e.g., gallium nitride high electron mobility transistors (GaN HEMTs), may be implemented in specific settings such as in power conversion circuits. III-V semiconductor materials thus have emerged as a leading technology for power switching, radio frequency (RF) and millimeter wave (mmWave) (e.g., 3-300 GHz) wireless applications. HEMTs offer high conduction and low resistive losses in comparison to conventional silicon-based devices.

(3) An electrically programmable fuse (efuse) is an electrical connection between two nodes that is configured to be destroyed in response to an electrical current, thus creating an open circuit between the two nodes. The effectiveness and reliability of efuses and other device components depend partially on each component's ability to resist or prevent intended degradation from sources other than electrical current, e.g., gradual electromigration degradation. In some cases, efuses may be difficult to integrate into circuits that have components with fundamentally distinct device architectures.

SUMMARY

(4) All aspects, examples and features mentioned herein can be combined in any technically possible way.

(5) Embodiments of a structure disclosed herein provide a structure including: a plurality of crystalline semiconductor layers, wherein each crystalline semiconductor layer includes a

compound material; a metallic layer on the plurality of crystalline semiconductor layers, wherein the metallic layer has a lower resistivity than an uppermost layer of the plurality of crystalline semiconductor layers; and a pair of gate conductors on respective portions of the metallic layer, wherein the metallic layer defines an electrically programmable fuse (efuse) link between the gate conductors.

(6) Some embodiments of a structure disclosed herein provide an electrically programmable fuse (efuse) structure for a III-V semiconductor substrate, the efuse structure including: a crystalline aluminum gallium nitride (AlGaN) layer at least partially on an isolation layer and above the III-V semiconductor substrate; a crystalline gallium nitride (GaN) layer on the crystalline AlGaN layer; a titanium nitride (TiN) layer on the crystalline GaN layer; and a pair of gate conductors on respective portions of the TiN layer, wherein the TiN layer defines an electrically programmable fuse (efuse) link between the gate conductors.

(7) Additional embodiments of a structure disclosed herein provide a method including: forming a plurality of crystalline semiconductor layers, wherein each crystalline semiconductor layer includes a compound material; forming a metallic layer on the plurality of crystalline semiconductor layers, wherein the metallic layer has a lower resistivity than an uppermost layer of the plurality of crystalline semiconductor layers; and forming a pair of gate conductors on respective portions of the metallic layer, wherein the metallic layer defines an electrically programmable fuse (efuse) link between the gate conductors.

(8) Another aspect of the disclosure includes any of the preceding aspects, and wherein a resistivity of the metallic layer is less than a resistivity of the uppermost layer of the plurality of III-V semiconductor layers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention will be better understood from the following detailed description with reference to the drawings, which are not necessarily drawn to scale and in which:

(2) FIG. 1 depicts a first cross-sectional view in plane X-Z of a structure according to embodiments of the disclosure.

(3) FIG. 2 depicts a second cross-sectional view in plane Y-Z of a structure, from view line 2-2 of FIG. 1, according to embodiments of the disclosure.

(4) FIG. 3 depicts a cross-sectional view in plane X-Z of a structure and adjacent transistor according to embodiments of the disclosure.

(5) FIG. 4 depicts a cross-sectional view in plane X-Z of a structure and adjacent transistor according to further embodiments of the disclosure.

(6) FIGS. 5-9 depict cross-sectional views in plane X-Z of methods to form a structure according to the disclosure.

DETAILED DESCRIPTION

(7) Integrated circuit (ICs) for implementing complex functions may include alternatingly doped semiconductor regions in various configurations. One type of configuration is a “triple well” structure, featuring a doped well instead of a deep well and various shallow wells inside of the well. These wells may have p-type and n-type doping, e.g., to provide opposite polarity metal oxide semiconductor field effect transistors (MOSFETs) in close proximity within a device layer. These features, while favorable for MOSFETs and similar devices, may present drawbacks when used for other types of devices. Electrically programmable fuses (“efuses”), in a triple well structure or similar configuration, require a significantly larger surface area to operate due to the wide variety of doping polarities and concentrations in a device. An efuse is an electrical connection between two nodes that is configured to degrade in response to an electrical current, thus creating an open

circuit between the two nodes. In triple well structures featuring efuses, the resistance of the open circuit may not be predictable and thus may pose reliability concerns.

(8) Embodiments of the disclosure provide an electrically programmable fuse structure over crystalline semiconductor layers, e.g., those used in III-V semiconductor structures such as high electron mobility transistors (HEMTs). Embodiments of the disclosure thus integrate efuse technology with structures featuring various types of transistors and/or other components where efuses have not previously been implemented. A structure according to the disclosure may include a plurality of crystalline semiconductor materials, each of which may include a compound material (e.g., a combination of semiconductor and metal material(s) as discussed herein). A metallic layer may be on the plurality of crystalline semiconductor materials and may feature a lower resistivity than an uppermost layer of the plurality of crystalline semiconductor layers. A pair of gate conductors may be on respective portions of the metallic layer, such that the metallic layer defines an efuse link between the gate conductors. During operation, transmitting a minimum current from one gate conductor to the other may destroy (i.e., “program”) the efuse, thus preventing further passage of current through the efuse structure.

(9) FIG. 1 shows a cross-sectional view of a structure **100** including an efuse **102** in a gallium nitride (GaN) technology setting, i.e., for an III-V integrated circuit, according to embodiments of the disclosure. Efuse **102** may be used in any III-V integrated circuit structure in which a selectively programmable conductive coupling finds application. Structure **100** and efuse **102** can be formed on a III-V semiconductor substrate, e.g., a gallium nitride (GaN) heterojunction structure including a barrier layer and a doped crystalline layer (e.g., configured for use as, or within, the channel layer of a transistor structure). More particularly, structure **100** and efuse **102** can include multiple epitaxially grown semiconductor layers on a semiconductor substrate **114**. Semiconductor substrate **114** can be, for example, a silicon or silicon-based substrate (e.g., a silicon carbide (SiC) substrate), a sapphire substrate, a III-V semiconductor substrate (e.g., a gallium nitride (GaN) substrate or some other suitable III-V semiconductor substrate), a silicon substrate (perhaps doped p-type), or any other suitable substrate for a III-V semiconductor device. Epitaxially grown semiconductor layers on substrate **114** can include, for example: a barrier layer **116** above the top surface of semiconductor substrate **114** and a doped crystalline layer **118** on barrier layer **116**. Barrier layer **116** in some cases may include aluminum gallium nitride (AlGaN) or any other material suitable for use as a barrier layer in a high electron mobility transistor. Where barrier layer **116** includes AlGaN, barrier layer **116** may have an Al mole fraction of between approximately fifteen percent and approximately twenty-five percent.

(10) An isolation layer **120** can vertically separate one or more portions of semiconductor substrate **114** and barrier layer **116** from each other. Isolation layer **120** may include, e.g., any currently known or later developed dielectric material (e.g., silicon dioxide (SiO₂) or any other suitable isolation material) capable of being formed as a layer at least partially on semiconductor substrate **114**. In some cases, isolation layer **120** may include various dopants, e.g., as an effect of doping semiconductor substrate **114** and/or any layers positioned thereover. In further implementations, isolation layer **120** may include one or more amorphizing dopant species (e.g., amorphous silicon (Si), argon (Ar), and/or other species configured to modify the crystal structure of semiconductor material). Such species may be embedded within isolation layer **120**, e.g., as a structural artifact of processes to form crystalline semiconductor layers such as barrier layer **116** and/or doped crystalline layer **118** thereover.

(11) In still further implementations, additional crystalline semiconductor layers may be on or included within semiconductor substrate **114**, barrier layer **116**, and/or doped crystalline layer **118**. For instance, one or more buffer layers (not shown) may be on semiconductor substrate **114** to enhance the forming of additional semiconductor material layers thereon. Where included, such buffer layers could be, for example, a gallium nitride (GaN) buffer layer, an aluminum nitride (AlN) buffer layer, or a buffer layer of any other material suitable for use as a buffer layer of a III-

V semiconductor device. Buffer layers, more specifically, can be employed to facilitate growth of barrier layer **116** and/or doped crystalline layer **118** and to provide for lattice constants of semiconductor substrate **114** below and doped crystalline layer **118** above. Buffer layers, where included, can be doped or undoped with any currently known or later developed dopant species (e.g., carbon doping).

(12) The crystalline semiconductor layers of structure **100** can be, for example, III-V semiconductor layers. Those skilled in the art will recognize that a III-V semiconductor refers to a compound obtained by combining group III elements, such as aluminum (Al), gallium (Ga), or indium (In), with group V elements, such as nitrogen (N), phosphorous (P), arsenic (As) or antimony (Sb)) (e.g., GaN, InP, GaAs, or GaP). Barrier layer **116** can have a band gap that is wider than the bandgap of doped crystalline layer **118**. As discussed in further detail elsewhere herein, such materials may be selected to form a heterojunction in desired locations, e.g., to produce two-dimensional electron gas (2DEG) regions for enabling the operation of a transistor structure (e.g., high electron mobility transistor (HEMT) structure) within structure **100**. In any case, the above-mentioned layers can be epitaxially grown by metal-organic chemical vapor deposition (MOCVD) or any other suitable technique (e.g., molecular beam epitaxy (MBE), hydride vapor phase epitaxy (HYPE), etc.). Furthermore, any one or more of the above-noted layers **116**, **118**, could be multi-layered structures (e.g., comprising multiple sub-layers of different buffer materials, multiple sub-layers of different III-V semiconductor channel materials and/or multiple sub-layers of different barrier materials).

(13) A metallic layer **122** including, e.g., titanium nitride (TiN) and/or any currently known or later developed semiconductor-metal composition, may be on doped crystalline layer **118**. Metallic layer **122** may extend horizontally between a pair of gate conductors **124**, each defining a respective cathode and anode terminal of efuse **102**. Although gate conductors **124** of efuse **102** do not function as transistor gates in efuse **102**, they may include one or more materials that are used elsewhere in structure **100**, and/or may be formed as part of the same process, as materials defining the gate terminal of one or more transistors as discussed elsewhere herein. As will also be discussed in further detail herein, metallic layer **122** may be substantially smaller in horizontal width than doped crystalline layer **118** and/or may also have a smaller width than gate conductor(s) **124**. A set of contacts **126** (e.g., metal wires or vias) on gate conductors **124** may electrically couple efuse **102** to other structures or devices, including a set of conductors **128**. Conductors **128** may include or otherwise may be coupled to wires and/or vias in overlying metal level layers (not shown).

(14) Metallic layer **122** may be structured such that current generally passes from one gate conductor **124** to another through metallic layer **122** without significant impedance. Although metallic layer **122** may include TiN as discussed herein, metallic layer **122** additionally or alternatively may include any composition having a resistivity that is less than that of doped crystalline layer **118** (or any other semiconductor layer) thereunder. Thus, metallic layer **122** conducts current from one gate conductor **124** to another, rather than directing a significant amount of current through underlying semiconductor material(s). Metallic layer **122**, additionally, may be entirely over isolation layer **120** to prevent any current therein from connecting to other structures through semiconductor substrate **114** and/or any crystalline semiconductor layers thereon. The passage of at least a minimum current through efuse structure **102**, however, may increase the thermal energy within metallic layer **122**, thereby destroying a portion of metallic layer **122** between gate conductors **124** when at least the minimum current passes therethrough. Metallic layer **122** thus defines a fuse link within efuse **102**, in that metallic layer **122** may be electrically destroyed to decouple gate conductors **124** from each other without affecting other portions of structure **100**. These properties of metallic layer **122** may arise from, e.g., metallic layer **122** being narrower in size yet longer in length than other portions of structure **100** having the same or similar composition. Thus, efuse structure **102** is capable of being programmed during the operation of structure **100**.

(15) Structure **100** and efuse **102** further may include one or more middle-of-the-line (MOL) dielectric layers **130**. The MOL dielectric layer(s) **130** can cover and, particularly, can have a bottom surface above and immediately adjacent to efuse **102** and its subcomponents, as well as any other features over semiconductor substrate **114**. MOL dielectric layer(s) **130** can include, for example, an etch stop material and a blanket interlayer dielectric (ILD) material layer on the etch stop layer. Etch stop materials can be, for example, a relatively thin conformal silicon nitride layer or a relatively thin conformal layer of some other suitable etch stop material. ILD materials can be, for example, a blanket layer of silicon dioxide, a doped silicon glass (e.g., phosphosilicate glass (PSG) or borophosphosilicate glass (BPSG)), or a blanket layer of any other suitable ILD material. The MOL dielectric layer(s) **130**, as illustrated, can fill the space that is above metallic layer **122** of efuse **102** that extends laterally between gate conductors **124**.

(16) Turning briefly to FIG. 2, illustrating a cross-section of structure **100** and efuse **102** along view line 2-2 of FIG. 1, additional features of efuse **102** are described. Gate conductor **124** may have a width $W_{sub.C}$ along the Y-axis direction that is greater than a width $W_{sub.M}$ of metallic layer **122** along the same direction. This difference in width, in some implementations, may cause gate conductor **124** to be connected to metallic layer **118** through its upper surface(s) and sidewall(s), thereby increasing the amount of contact area between these two elements. In addition, metallic layer **122** having a narrower width along the Y-axis than gate conductor **124** may reduce the minimum current needed to destroy metallic layer **122** and thereby program efuse **102**. The difference between widths $W_{sub.C}$, $W_{sub.M}$ may vary based on the intended application and/or technical setting of structure **100**. It is understood that in further embodiments or implementations, width $W_{sub.C}$ may be less than or approximately equal to width $W_{sub.M}$.

(17) Turning to FIG. 3, a cross-sectional view of structure **100** is shown according to further examples. Here, structure **100** includes efuse **102** and a transistor structure **150** adjacent efuse **102** over semiconductor substrate **114**. Transistor structure **150** may be provided in the form of a high electron mobility transistor (HEMT) structure and/or other types of transistor structures capable of being formed on III-V semiconductor materials such as GaN or other semiconductor substrate **114** compositions discussed herein. In transistor structure **150**, the composition of semiconductor substrate **114** and barrier layer **116** can be selected so that a heterojunction is formed at the interface between the two layers, thereby forming a two-dimensional electron gas (2DEG) region **152** in semiconductor substrate **114** (see dashed box). 2DEG region **152** in semiconductor substrate **114** can provide the conductive pathway for the drifting of charges between the source and the drain suitable for use in transistor structure **150** or, in more specific examples, an HEMT or metal-insulator-semiconductor HEMT (MISHEMT). Doped crystalline layer **118** may be on barrier layer **116** in transistor structure **150**. Collectively, doped crystalline layer **118** and barrier layer **116** form a gallium nitride (GaN) heterojunction structure for operating transistor structure **150**. Metallic layer **122** may be on doped crystalline layer **118** in transistor structure **150** but defines a vertical electrical coupling. Transistor structure **150** also may include gate conductor **124** coupled to barrier layer **116** through doped crystalline layer **118** and metallic layer **122**. A gate contact **154** to gate conductor **124** defines a gate terminal for control of current flow through transistor structure **150**. Metallic layer **122** will not operate as a fuse because it only has a single gate contact **154** thereto, and thus current does not pass laterally through metallic layer **122** in transistor structure **150**.

(18) Transistor structure **150** may be implemented in a variety of forms. A depletion-mode transistor refers to a device that allows current to pass from source to drain when the gate voltage is less than its pinch-off voltage but prevents current from passing from source to drain when the gate voltage meets or exceeds the pinch-off voltage. By contrast, an enhancement-mode transistor refers to a device that prevents current from passing from source to drain when the gate voltage is less than its threshold voltage (i.e., distinguishable from “pinch-off voltage” in the case of a depletion-mode transistor) but allows current flow from source to drain when the gate voltage meets or exceeds the transistor's threshold voltage. Embodiments of the disclosure integrate efuse **102** with

transistor structure **150** regardless of whether transistor structure **150** includes depletion-mode (DMODE) or enhancement-mode structures (EMODE) therein. In the case of an EMODE device, transistor structure **150** may not include doped crystalline layer **118** and metallic layer **122** therein. Thus, FIG. **3** depicts transistor structure **150** by example as being a DMODE device.

(19) In still further implementations, efuse **102** and/or transistor structure **150** may have distinct materials within barrier layer **116** apart from AlGaN to further distinguish their response to voltage biases during operation. In the case where gate barrier layer **116** includes AlGaN, additional doping of materials within transistor structure **150** may not be necessary, e.g., due to the intrinsic n-type doping of GaN. In this case, gate conductor **124** may have the same composition as any conductive contacts for source and/or drain contacts to transistor structure **150**. In the case where transistor structure **150** is an enhancement-mode transistor, gate conductor **124** additionally may include doped crystalline layer **118** formed of pGaN. The presence of pGaN within doped crystalline layer **118** may cause transistor structure **150** to operate as an enhancement-mode transistor.

(20) The electrical biasing of transistor structure **150** at gate contact **154** can affect whether current may flow between a set of source/drain terminals **160** of transistor structure **150**. Source/drain terminals **160** are labeled separately as **160a**, **160b** in the accompanying FIGS. to indicate that either terminal may function as a source or drain of transistor structure **150**. To define source/drain terminals **160a**, **160b**, structure **100** may include a set of ohmic contacts **162** on opposite horizontal ends of barrier layer **116**, thus providing a physical interface between ohmic contacts **162** and 2DEG region **152**. Ohmic contacts **162** include one or more metals configured for electrically interfacing with semiconductor materials(s) to define a metal-to-semiconductor Schottky diode within structure **100**. Ohmic contacts **162** may be formed, e.g., by surface cleaning of semiconductor substrate **114**, forming metal(s) on the cleaned surface, patterning, and annealing to create ohmic contacts **162** in desired positions. In the case where semiconductor substrate **114** includes GaN, ohmic contacts **162** may include composite materials of titanium (Ti), aluminum (Al), nickel (Ni), and gold (Au), or in another example may include palladium (Pd)/Au material. A set of source/drain (S/D) contacts **164** (e.g., metal wires or vias) on ohmic contacts **162** may electrically couple transistor structure **150** to other structures or devices, e.g., any of conductors **128**. Conductors **128**, in some cases, may couple one source/drain terminal **160a**, **160b** to the cathode or anode terminal of efuse **102**.

(21) Turning briefly to FIG. **4**, isolation layer **120** may be provided in a variety of shapes and positions in further embodiments. Although isolation layer **120** may extend entirely beneath metallic layer **122** in some embodiments (e.g., as shown in FIGS. **1-3**) to prevent conductive pathways from forming through semiconductor substrate **114** below efuse **102**, this is not necessarily required. In some cases, multiple isolation layers **120a**, **120b**, **120c** can be within semiconductor substrate **114** to horizontally separate portions of semiconductor substrate **114** below efuse **102** from portions of semiconductor substrate **114** below transistor structure **150**. Even where isolation layer(s) **120a**, **120b**, **120c** extend only partially into semiconductor substrate **114**, such a depth may be sufficient to stop current from flowing from efuse **102** to active portions of transistor structure **150**, e.g., 2DEG region **152**. In this case, a lower surface of barrier layer **116** below metallic layer **122** may physically interface with semiconductor substrate **114** without isolation layer **120** therebetween.

(22) Turning now to FIG. **5**, embodiments of the disclosure include processing methodologies to form an electrically programmable fuse, e.g., efuse **102**, within embodiments of structure **100**. A preliminary structure **170** may include semiconductor substrate **114**, with barrier layer **116** thereover as a single layer. Doped crystalline layer **118** may be on barrier layer **116** and metallic layer **122** may be on barrier layer **118**. Notably, in some embodiments, isolation layer **120** (FIGS. **1-4**) may not be initially present within preliminary structure **170**. Layers **116**, **118**, **122** on semiconductor substrate **114** may be stacked such that there are no structural differences in preliminary structure **170** where efuse **102** (FIGS. **1-4**) and transistor structure **150** (FIGS. **3**, **4**)

will be formed.

(23) Referring to FIG. 6 methods of the disclosure may include forming isolation layer **120** on semiconductor substrate **114** but underneath barrier layer **116**, even after doped crystalline layer **118** and metallic layer **122** have been formed. Isolation layer **120** may be formed, e.g., by implanting insulative materials (e.g., Nitrogen, Argon, and/or other similar materials currently known or later developed) such that they pass through layers **116**, **118**, **122** to enter the uppermost portions of semiconductor substrate **114**. In the case where isolation layer **120** is formed by implantation of insulative materials, the implanted materials may convert some upper portions of semiconductor substrate **114** into isolation layer **120**. As discussed elsewhere herein, the forming of isolation layer **120** may be implemented with the aid of masks and/or other patterning materials to form multiple isolation layers (e.g., isolation layers **120a**, **120b**, **120c** shown in FIG. 4) on semiconductor substrate **114**, and/or to prevent isolation layer **120** from being formed in locations where 2DEG region **152** (FIGS. 3, 4) is desired. In alternative implementations, isolation layer **120** may be formed before any of layers **116**, **118**, **122** are formed on semiconductor substrate **114**.

(24) FIG. 7 depicts further processing to define portions of efuse **102** (FIGS. 1-4) and transistor structure **150** (FIGS. 3, 4) over semiconductor substrate **114**. To further define the positions of efuse **102** and transistor structure **150**, portions of layers **118**, **122** can be removed such that two or more distinct layers **118**, **122** are over barrier layer **116**. The processing of layers **118**, **122** may include, e.g., recessing TiN in metallic layer **122** and selectively etching the composition of doped crystalline layer **118** (e.g., pGaN). Ohmic contacts **162** can be formed on and within barrier layer **116**, e.g., by removing portions of barrier layer **116** horizontally between layers **118**, **122** and forming ohmic metals over isolation layer **120**. In addition, a portion of MOL dielectric layer(s) **130** may be formed over layers **118**, **120**, **122** and ohmic contacts **162**. MOL dielectric layer(s) **130** also may be planarized to yield a planar upper surface but may nonetheless cover metallic layer(s) **122**, crystalline semiconductor layer(s) **118**, and ohmic contacts **162**.

(25) FIG. 8 depicts forming a gate conductor layer **172**, e.g., by forming openings in MOL dielectric **130** to expose portions of metallic layers **122** and depositing conductive gate material(s) therein. The openings in MOL dielectric **130** over metallic layers **122** may be formed by targeted etching of MOL dielectric **130**. Gate conductor layer **172** may have the same composition as gate conductor(s) **124** (FIGS. 1-4) discussed elsewhere herein. Upon being formed, gate conductor layer **172** may fill the openings over metallic layers **122** and cover MOL dielectric **130**. Gate conductor layer **172** can be processed subsequently to define the cathode and anode of efuse **102** (FIGS. 1-4) and gate terminal of transistor structure **150** (FIGS. 3, 4).

(26) FIG. 9 depicts the processing of gate conductor layer **172** (FIG. 8), e.g., by selective etching of gate conductor material, to yield gate conductors **124** over each metallic layer **122**. As shown in FIG. 9, an upper portion of each gate conductor **124** may overhang a lower portion of gate conductor **124** thereunder, e.g., due to the partial forming of gate conductor layer **172** on MOL dielectric **130**. Additional portions of MOL dielectric **130** then may be formed to a desired height, thus covering gate conductors **124**. At this stage of processing, additional openings may be formed within MOL dielectric **130** to form metal wires and vias, e.g., contacts **126** (FIGS. 1, 3, 4), conductors **128** (FIGS. 1, 3, 4), gate conductor(s) **154** (FIGS. 3, 4), and/or S/D contact(s) **164** (FIGS. 3, 4). Such processing may yield, e.g., structure **100** with the various structural features discussed herein, and including efuse **102** therein.

(27) Embodiments of the disclosure provide various technical and commercial advantages, examples of which are discussed herein. Various devices having HEMTs therein, particularly those integrating GaN into power generation or amplification components, may be intended for use in various settings where efuses are desirable. As examples, circuitry to detect excessive temperatures may rely on efuses to cease operation of the circuit when temperature exceeds a threshold level, or programmable resistors may use efuses to encode logical functions prescribed by the user(s). Due to the need for such circuitry in power generation and amplification circuits, embodiments of the

disclosure allow efuse hardware to be structurally integrated into the same portions of a device as HEMT structures and/or similar III-V device architectures without varying the type of semiconductor material(s) needed to provide these devices.

(28) In the description of the structure embodiments above, reference is made to semiconductor features including, but not limited to, substrates, layers, regions, portions, terminals, etc. and to the conductivity types (e.g., P-type or N-type) of those features. Such features are made up of one or more semiconductor materials and a semiconductor material refers to a material whose conducting properties can be altered by doping with an impurity. Illustrative semiconductor materials include, for example, silicon-based semiconductor materials (e.g., silicon, silicon germanium, silicon germanium carbide, silicon carbide, etc.) and III-V compound semiconductors (i.e., compounds obtained by combining group III elements, such as aluminum (Al), gallium (Ga), or indium (In), with group V elements, such as nitrogen (N), phosphorous (P), arsenic (As) or antimony (Sb)) (e.g., GaN, InP, GaAs, or GaP). A pure semiconductor material and, more particularly, a semiconductor material that is not doped with an impurity for the purposes of increasing P-type or N-type conductivity is referred to in the art as an intrinsic semiconductor (i.e., an undoped semiconductor). A semiconductor material that is doped with an impurity for the purposes of increasing P-type or N-type conductivity is referred to in the art as an extrinsic semiconductor (i.e., a doped semiconductor). Such an extrinsic semiconductor will be P-type (i.e., will have P-type conductivity) or will be N-type (i.e., will have N-type conductivity) and will be more conductive than an intrinsic semiconductor made of the same base material. That is, extrinsic silicon will be more conductive than intrinsic silicon; extrinsic silicon germanium will be more conductive than intrinsic silicon germanium; and so on. Furthermore, different impurities (i.e., different dopants) can be used to achieve the different conductivity types (e.g., P-type conductivity and N-type conductivity) and the dopants may vary depending upon the different semiconductor materials used. For example, a silicon-based semiconductor material (e.g., silicon, silicon germanium, etc.) is typically doped with a Group III dopant, such as boron (B) or indium (In), to achieve P-type conductivity, whereas a silicon-based semiconductor material is typically doped a Group V dopant, such as arsenic (As), phosphorous (P) or antimony (Sb), to achieve N-type conductivity. A gallium nitride (GaN)-based semiconductor material is typically doped with magnesium (Mg) to achieve P-type conductivity and with silicon (Si) or oxygen to achieve N-type conductivity. Those skilled in the art will also recognize that different conductivity levels will depend upon the relative concentration levels of the dopant(s) in each semiconductor region.

(29) The terminology used herein is for the purpose of describing the disclosed structures and methods and is not intended to be limiting. For example, as used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Additionally, as used herein, the terms “comprises,” “comprising,” “includes,” and/or “including” specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Furthermore, as used herein, terms such as “right,” “left,” “vertical,” “horizontal,” “top,” “bottom,” “upper,” “lower,” “under,” “below,” “underlying,” “over,” “overlying,” “parallel,” “perpendicular,” etc., are intended to describe relative locations as they are oriented and illustrated in the drawings (unless otherwise indicated) and terms such as “touching,” “in direct contact,” “abutting,” “directly adjacent to,” “immediately adjacent to,” etc., are intended to indicate that at least one element physically contacts another element (without other elements separating the described elements). The term “laterally” is used herein to describe the relative locations of elements and, more particularly, to indicate that an element is positioned to the side of another element as opposed to above or below the other element, as those elements are oriented and illustrated in the drawings. For example, an element that is positioned laterally adjacent to another element will be beside the other element, an element that is positioned laterally immediately adjacent to another element will be directly beside

the other element, and an element that laterally surrounds another element will be adjacent to and border the outer sidewalls of the other element. The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed.

(30) The descriptions of the various embodiments of the present invention have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A structure comprising: a plurality of crystalline semiconductor layers, wherein each crystalline semiconductor layer includes a compound material; a metallic layer on the plurality of crystalline semiconductor layers, wherein the metallic layer has a lower resistivity than an uppermost layer of the plurality of crystalline semiconductor layers; and a pair of gate conductors on respective portions of the metallic layer, wherein the metallic layer defines an electrically programmable fuse (efuse) link between the gate conductors.
2. The structure of claim 1, wherein the plurality of crystalline semiconductor layers includes: an isolation layer on a portion of a gallium nitride (GaN) layer; an aluminum gallium nitride (AlGaN) layer on one of the isolation layer and the GaN layer; and a p-type gallium nitride (pGaN) layer on the AlGaN layer, wherein the pGaN layer is the uppermost layer of the plurality of crystalline semiconductor layers.
3. The structure of claim 2, wherein the GaN layer and the AlGaN layer are included within an adjacent transistor structure on the GaN layer.
4. The structure of claim 3, further comprising a conductor coupling one of the pair of gate conductors to a source/drain terminal of the adjacent transistor structure.
5. The structure of claim 2, wherein the isolation layer includes an amorphizing dopant species.
6. The structure of claim 1, wherein a resistivity of the metallic layer is less than a resistivity of the uppermost layer of the plurality of crystalline semiconductor layers.
7. The structure of claim 1, wherein the metallic layer includes titanium nitride (TiN).
8. An electrically programmable fuse (efuse) structure for a III-V semiconductor substrate, the efuse structure comprising: a crystalline aluminum gallium nitride (AlGaN) layer at least partially on an isolation layer and above the III-V semiconductor substrate; a crystalline gallium nitride (GaN) layer on the crystalline AlGaN layer; a titanium nitride (TiN) layer on the crystalline GaN layer; and a pair of gate conductors on respective portions of the TiN layer, wherein the TiN layer defines an electrically programmable fuse (efuse) link between the gate conductors.
9. The efuse structure of claim 8, wherein the III-V semiconductor substrate includes GaN.
10. The efuse structure of claim 8, wherein the AlGaN layer has an Al mole fraction of between approximately 15% and approximately 25%.
11. The efuse structure of claim 8, wherein the isolation layer includes an argon (Ar) dopant.
12. The efuse structure of claim 8, wherein the TiN layer is entirely above the isolation layer.
13. The efuse structure of claim 8, wherein a lower surface of the crystalline AlGaN layer includes an interface with the III-V substrate that does not include the isolation layer therebetween.
14. The efuse structure of claim 8, further comprising a transistor structure over the crystalline AlGaN layer and horizontally adjacent the TiN layer.
15. A method comprising: forming a plurality of crystalline semiconductor layers, wherein each

crystalline semiconductor layer includes a compound material; forming a metallic layer on the plurality of crystalline semiconductor layers, wherein the metallic layer has a lower resistivity than an uppermost layer of the plurality of crystalline semiconductor layers; and forming a pair of gate conductors on respective portions of the metallic layer, wherein the metallic layer defines an electrically programmable fuse (efuse) link between the gate conductors.

16. The method of claim 15, wherein forming the plurality of crystalline semiconductor layers includes: forming an isolation layer on a portion of a gallium nitride (GaN) layer; forming an aluminum gallium nitride (AlGaN) layer on one of the isolation layer and the GaN layer; and forming a p-type gallium nitride (pGaN) layer on the AlGaN layer, wherein the pGaN layer is the uppermost layer of the plurality of crystalline semiconductor layers.

17. The method of claim 16, further comprising forming an adjacent transistor structure on the GaN layer, wherein the GaN layer and the AlGaN layer are included within the adjacent transistor structure.

18. The method of claim 16, wherein forming the metallic layer causes the metallic layer to be entirely above the isolation layer.

19. The method of claim 16, wherein forming the isolation layer includes introducing an amorphizing dopant species into the isolation layer.

20. The method of claim 15, wherein the metallic layer includes titanium nitride (TiN).
